# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A lunar eclipse can occur only on: 1

**(a)** any day **(b)** a new moon day **(c)** first quarter **(d)** a full moon day

**2.** Between 21 March and 21 June, the length of day in the northern hemisphere: 1

**(a)** becomes zero **(b)** decreases **(c)** increases **(d)** stays same

**3.** The stars appear to trace circles around the Pole Star in: 1

**(a)** one year **(b)** one month

**(c)** one week **(d)** one day (24 hours roughly)

**4.** A lunar eclipse is visible from: 1

**(a)** the whole night side of the Earth **(b)** only the equator

**(c)** only the poles **(d)** a narrow strip only

**5.** After 21 June, the noon Sun's elevation at a place in the northern hemisphere: 1

**(a)** increases **(b)** becomes zero **(c)** decreases **(d)** stays same

**6.** The same constellation rises about 4 minutes earlier every night because: 1

**(a)** clouds **(b)** stars move

**(c)** the Moon moves **(d)** Earth revolves around the Sun

**7.** The umbra of the Moon's shadow on the Earth causes: 1

**(a)** tides **(b)** a total solar eclipse

**(c)** a lunar eclipse **(d)** seasons

**8.** Time taken by the Moon to complete one rotation on its axis is equal to: 1

**(a)** 365 days

**(b)** 7 days

**(c)** 24 hours

**(d)** its time period of revolution around the Earth

**9.** The summer constellations differ from winter ones because the Earth: 1

**(a)** tilts to the left **(b)** changes its position in its orbit

**(c)** stops rotating **(d)** moves away from stars

**10.** It is safe to see a total solar eclipse: 1

**(a)** only with proper filters **(b)** through a camera lens without filter

**(c)** with the naked eye **(d)** through sunglasses

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Why are festivals like Diwali on different dates every year in the Gregorian calendar? 2

**12.** How many leap years are there between 2000 and 2020, including 2000 and 2020? 2

**13.** Why do we not see an eclipse every month? 2

**14.** Name two instruments of Jantar Mantar and state their function. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Why does the Moon show the same face always? Explain the effect with a walking demonstration. 3

**16.** Describe an activity with a torch, a ball and a lamp to show phases of the Moon. 3

**17.** A calendar has 12 months of 30 days. How many days short is it of the solar year? Explain how the real calendar corrects it. 3

OR

Explain the 365.25-day solar year and the leap year rule with a calculation. 3

**18.** Why does the Sun's path differ in summer and in winter? 3

OR

Explain why a lunar eclipse can be seen from anywhere on the night side of the Earth but a solar eclipse only in a narrow region. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Describe how solar and lunar eclipses occur with positions and precautions for viewing. 5

OR

Explain how to use a simple model of a lamp (Sun), a globe (Earth) and a ball (Moon) to show day-night, seasons, phases and eclipses. 5

**20.** Explain the use of Jantar Mantar instruments and the contribution of Indian astronomers to timekeeping. 5

OR

Discuss the effects of changing latitudes on day length and the position of the Sun at noon on 21 June and on 22 December. 5

## END OF QUESTION PAPER — Total: 40 Marks

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